7 ADDITIONAL STUDIES

7.1 RISK ASSESSMENT

7.1.1 Scope

Scope of the risk assessment covers the storage of all chemicals involved in proposed facilities (as required in ***ToR 39***) and includes:

- Consequence Analysis for worst case scenario and maximum credible scenarios
- Quantitative Risk Analysis providing results for individual risk and societal risk
- Recommendation for risk reduction methods

7.1.2 Methodology

Risk is the combination of severity of consequence and the likelihood of occurrence. Some failure events may have potential for consequences of high severity but the likelihood of occurrence would be very low. On the other hand some failure events with low severity consequence might occur very frequently.

Risk = (Severity of Consequence) × (Likelihood of Occurrence)

Quantitative risk assessment (QRA) provides measure of risk in numerical value for specified time period, which is normally one year.

The five normal components of a QRA study are:

- Hazard, (or failure case) identification
- Failure frequency estimation
- Consequence calculations
- Risk calculation (Risk Summation)
- Risk assessment (using appropriate risk acceptability criteria)

The objective of risk assessment is to bring the risk 'as low as reasonably practicable (ALARP)'.

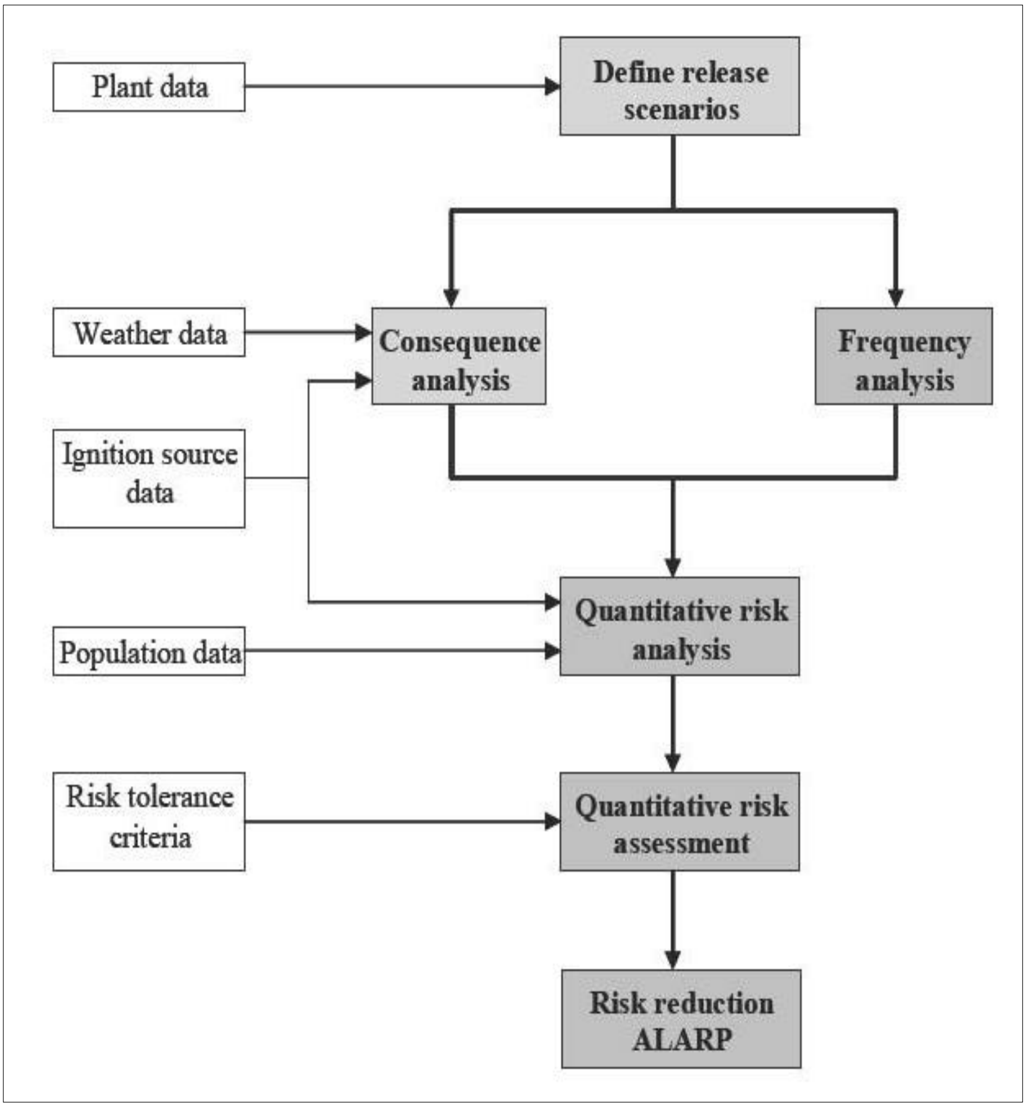
The risk levels associated with the facilities are presented in the following standard forms:

- **Individual risk contours** which show the geographical distribution of risk to an individual.
- **Group risk (FN) curves** which show the cumulative frequency (F) distribution of accidents causing different numbers (N) of fatalities. The FN curve therefore indicates whether the societal risk to the facility is dominated by relatively frequent accidents causing small numbers of fatalities or low frequency accidents causing many fatalities.

Approach to QRA (process chart)

The steps involved in preparation of QRA study are presented in ***Figure 7-1***

Figure 7-1: QRA process chart



7.1.3 Hazard Identification

The following hazards are identified for the purpose of this QRA study:

- Chlorine storage tanks
- Hydrogen storages – gas holder and cylinder banks
- Fuel oil and HSD storage tanks

Failure scenarios

In this study, toxic and flammable hazards are relevant. There is a possibility of failure associated with each mechanical component of the plant (vessels, pipes, pumps or compressors). These are generic failures and can be caused by such mechanisms as corrosion, vibration or external impact (mechanical or overpressure).

The range of possible releases for a given component covers a wide spectrum, from a pinhole leak up to a catastrophic rupture (of a vessel) or full bore rupture (of a pipe). For the purpose of QRA in an objective manner, representative failure cases are generated covering both the range of possible releases and their total frequency.

Accordingly, the following typical types of failures are considered:

- Minor Leak: 5mm
- Medium Leak: 25mm
- Large leak: 100 mm
- Tank – Catastrophic Rupture

For each identified failure case, the appropriate data required to define that case is input to the past risk package.

Input data

The input data required for QRA consist of the following:

- Plant data including process parameters, inventory of hazardous material in the section, equipment layout drawing and site location map
- Generic failure rate data for the type of equipment covered in the QRA study
- Meteorological data including ambient temperature, humidity and wind rose data
- Ignition source data including electrical installations, fired heaters, roads etc
- Population data including distribution of people in and around the plant

Software

DNV risk analysis software package **Phast version 7.1** is used in this QRA study.

Phast has been in use worldwide more than 20 years. It is based on the well validated Unified Dispersion Model (UDM) version 2.

Consequence Analysis for selected failure scenarios

Consequence analysis using Phast software has been carried out to provide results for the following:

- Toxic dispersion of chlorine
- Jet fire for hydrogen gas
- Vapor cloud explosion for hydrogen gas
- Pool (dyke) fire in case of fuel oil and HSD tanks

Results for the consequence analysis calculations are provided in the subsequent sections.

7.1.4 Input data for Risk Assessment

Plant Data

Details of storage systems having large inventory of hazardous materials relevant for this study are listed in ***Table 7-1***

Table 7-1: Details of Storage system of Hazardous material

Description	Nos.	Capacity (Each)	Capacity (Total)	Pressure	Temperature
Liquid Chlorine Tanks	4	150 MT	600 MT	8 bar(g)	30 °C
Filled Chlorine Tonners	1500	0.9 MT	1350 MT	8 bar(g)	30 °C
Hydrogen Gas Holder	1	250 m^3	250m^3	200 bar(g)	30 °C
HSD Tank	1	20 KL	20 KL	Atmospheric	30 °C

Input data for risk analysis including the process parameters and failure rates for defined scenarios are shown in ***Table 7-2***

Table 7-2: Input data for risk analysis

S.No	Description	Material & Phase	Temp.	Pressure	Leak size
1.	Chlorine Tanks	Liq. Chlorine	30°C	8 bar(g)	25 mm
2.	Chlorine Tonners	Liq. Chlorine	30°C	8 bar(g)	15 mm
3.	Hydrogen Gas Holder	Hydrogen Gas	30°C	200 mm WG	15 mm
4.	HSD Tank	HSD	30°C	Atmospheric	Rupture

Ignition Sources

Location of the ignition sources is significant for estimating fire and explosion effects in case of release of flammable materials such as hydrogen.

Common ignition sources are following:

- Fire heaters;
- Vehicles on roads;
- Electrical installations (switch gear room, transformer yard, DG room etc.);
- Overhead HT power lines.

7.1.5 Consequence Analysis Results

The results of consequence analysis for selected failure scenarios are summarized in ***Table 7-3***

Table 7-3: Results of Consequence analysis

Description		Down Wind Distance (m)		
Weather Parameters →		**E; 2.9 m/s**	**D; 3.1 m/s**	**D; 3 m/s**
1. Chlorine Tank– 25 mm Leak				
Toxic Dispersion of Chlorine	IDLH: 10 ppm	2,576	1,599	1,452
2. Chlorine Tonner Tube Failure				
Toxic Dispersion of Chlorine	IDLH: 10 ppm	1,406	874	815
3. Hydrogen Cylinder Bank				
Jet fire radiation intensity	4 kW/m2	37	37	37
	12.5 kW/m2	29	28	29
	37.5 kW/m2	22	22	24
VCE Overpressure	0.02 bar (0.3 psi)	101	103	99
	0.07 bar (1 psi)	59	60	58
	0.21 bar (3 psi)	44	45	44
4. HSD Tank – Catastrophic Rupture				
Late Ignition	0.02 bar	829	792	764
	0.07 bar	441	420	396
	0.21 bar	302	287	265

Figure 7-2: Maximum Concentration foot print due to 25mm leak in Chlorine tank at weather condition 3/D

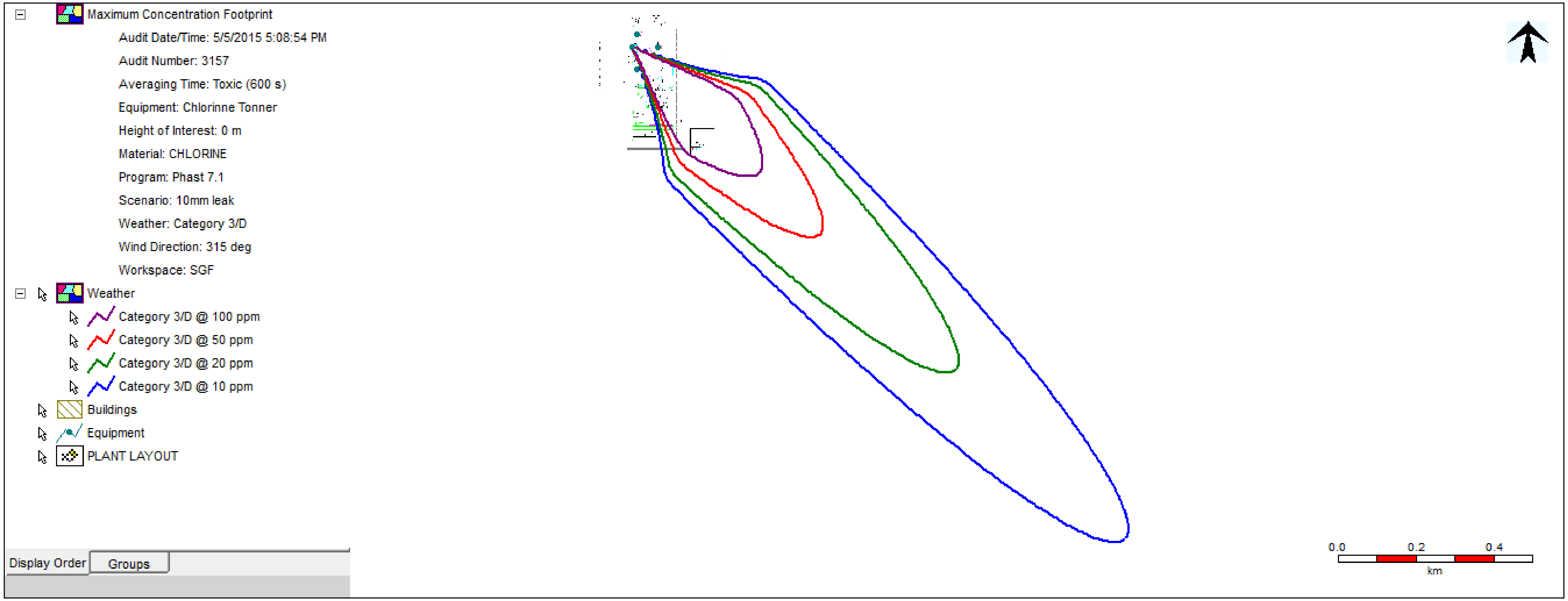


Figure 7-3: Late Explosion effect contour due to 15 mm leak in Hydrogen cylinder at weather condition 3/D

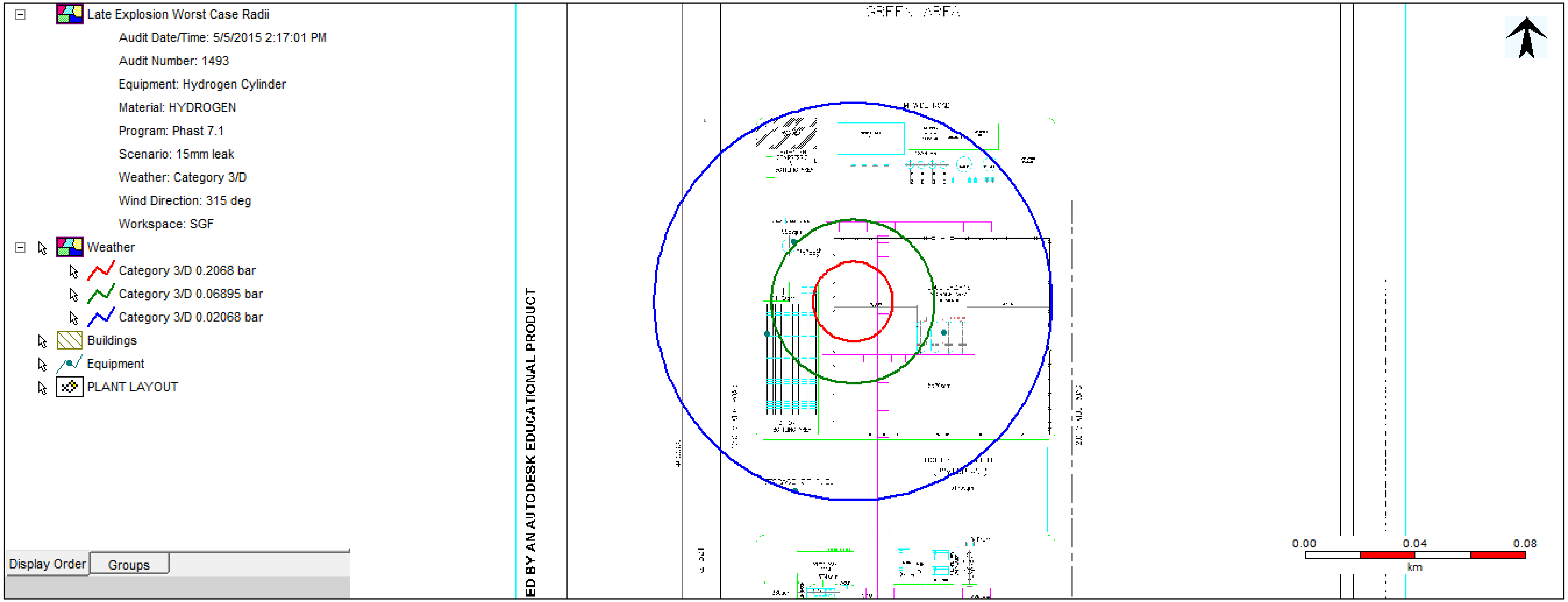


Figure 7-4: Jet fire effect due to 15mm leak in Hydrogen tcylinder at weather condition 3/D

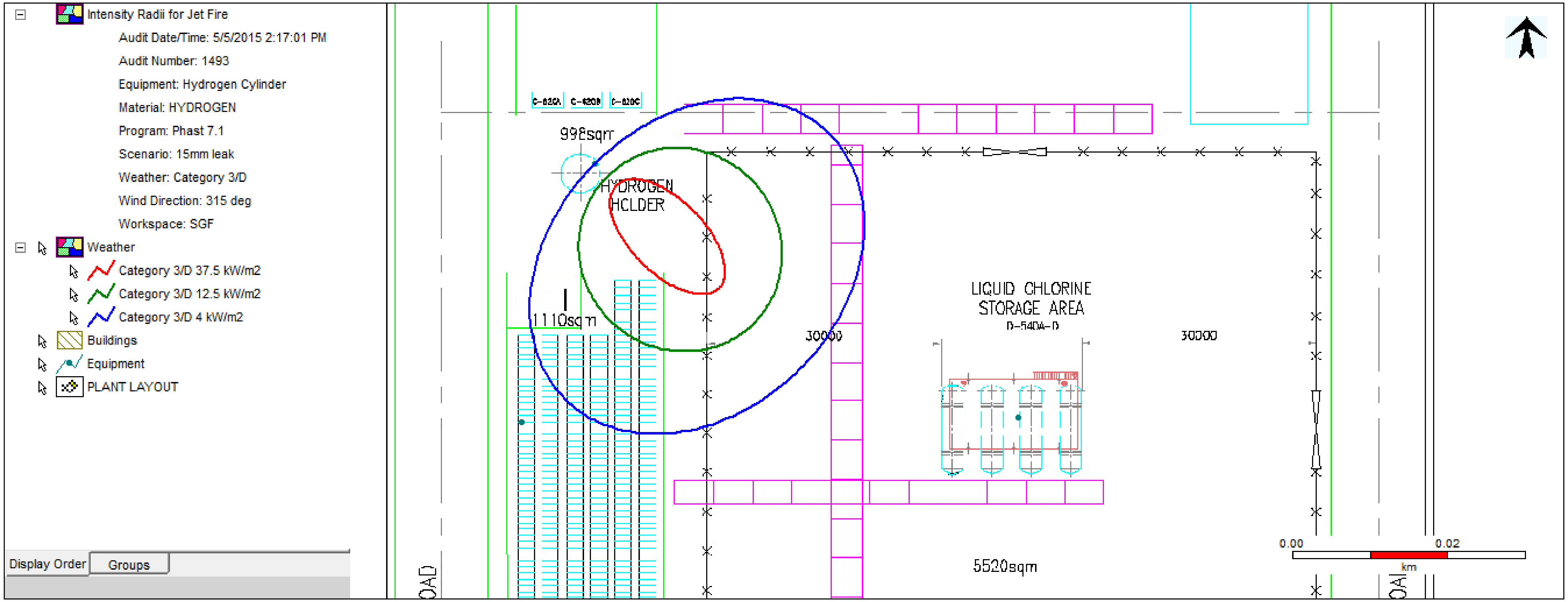
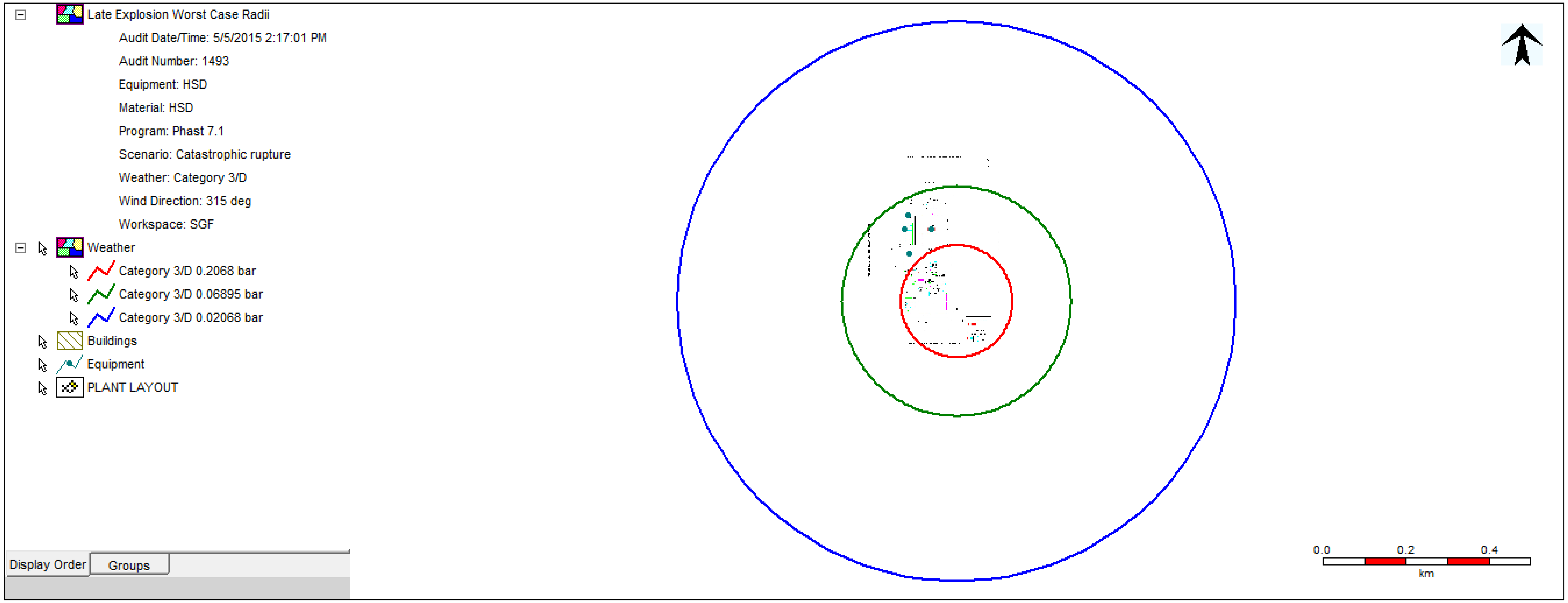


Figure 7-5: Late explosion overpressure effect due to catastrophic rupture of HSD storage tank at weather condition 3/D



7.1.6 Action Plan for Risk Reduction

Salient safety measures will be provided in the plant system and recommendations for improvement are indicated below.

1. Storage systems holding large inventories of liquid chlorine and hydrogen are the major risk contributors.
5. SGF have to maintain highly effective process safety management system.
6. Mechanical integrity assurance programme is most important to prevent loss of containment incidents involving hazardous materials. The following safety measures will be implemented in the system.
    - Chlorine tanks shall be subjected to rigorous periodic testing and inspection.
    - One tank shall always be kept empty to facilitate transfer from another tank in case of emergency.
    - All connections to chlorine tank shall be provided at the top and no pipe shall be connected to the tank bottom.
    - Liquid chlorine connection to the tank top shall be provided with emergency isolation valve.
7. Timely detection of leaks is critical to prevent major incidents. Chlorine leak detectors shall be provided in facility.
8. Chlorine leak absorption system with adequate capacity shall be provided at facility to absorb chlorine gas in circulating caustic soda solution to produce hypochlorite.
9. If hypochlorite solution gets inadvertently mixed with acid, there will be chlorine gas evolution at ground level and this will affect persons nearby. Care shall be taken to segregate the areas in which hypochlorite and acid are handled.

7.1.7 Emergency Organization

Emergency Response Team

The Operational Matrix shows the flow of information in an emergency and Organogram of designated persons constituting the Emergency Response Team.

Incident Informer

An Incident Informer is a person who on noticing an abnormal or hazardous situation in the plant immediately communicates the same to the Shift Engineer. The Shift Engineer on receiving the information immediately contacts the Incident Controller. There may be situations when the Shift Engineer himself anticipates an emergency from the control room itself from the instruments and alarms.

Incident Controller

He is a designated authority responsible for assessing and declaring an emergency, and in directing emergency operations in the affected area of the plant. He is fully familiar with the plant facilities, its operations and personnel. Till his arrival at the incident site, the Shift Engineer acts on his behalf as an alternate and initiates necessary emergency actions, and there after assists him in implementing emergency procedures.

Site Main Controller

He is a designated authority heading the Emergency Response Team and is also familiar with the plant and the personnel and is responsible for assessing and declaring the emergency as well as for calling off the emergency.

On being informed about the emergency in the plant, he proceeds to the Emergency Control Centre and on arrival at the plant assumes overall control of the emergency. Till his arrival, the Incident Controller acts on his behalf.

Based on the magnitude of the emergency, the Site Main Controller in consultation with the Incident Controller may requisition any or all of the following members of the team to report to him for further assistance.

Emergency Services

- In-charge (Environment & Safety)
- In-charge (Admin, Security & Fire)
- Male nurse/ Medical officer / trained first aiders / Trained fire warden.
- Security Officer / Fire Men / Security Guards

Support Services

- HOD- Power Plant / S/I – Power plant
- HOD - Mechanical
- HOD - Electrical /S/I - Electrical
- HOD - Instruments
- HOD - HRD

Responsibilities of Emergency Personnel

1. Site Main Controllers

- Site Main Controller relieves the Incident Controller of the responsibilities of overall emergency control as soon as he arrives in the plant and takes stock of the situation and thereafter will position himself in the Emergency Control Centre and give directions from there.
- Exercise direct control over areas of the plant which are affected by the incident and constantly review the situation in the affected area with the Incident Controller.
- Assesses the magnitude of the incident and decides to call additional personnel and to shut down the total operations in the plant and loading unloading operations depending on the nature of emergency.
- Ensure that affected personnel are transported to external medical centres, if required and keep constant liaison with these medical centers during the course of the emergency through the medical officer.
- Keeps Corporate Office and concerned Government Agencies informed of the emergency and if necessary arranges information to the outside habitants through police.
- Decides to call off the emergency.

2. Incident controllers

- Incident Controller on reaching the site of the incident relieves the Shift Engineer of the responsibilities of directing the emergency operations and assumes total control of emergency operations in the affected area.
- Assesses the magnitude of the incident and decides to call the emergency.
- Directs emergency operations from the incident site keeping in mind the priorities for safety of personnel, least damage to the property and environment and minimum loss of materials.
- Ensures that all non-essential personnel of the affected area are evacuated to the appropriate assembly points for evacuation to a safer place.
- Provide advice and information to the Fire and Security Personnel and Local Fire Service as and when they are called.
- Ensure removal the casualties to the medical center and if required arranges for taking a head count.
- Establishes contact and maintains communications with the Emergency Control Centre, and continuously reviews the situation with the Site Main Controller.
- To arranges head count at assembly point with the help of senior most person available at the assembly point.

10. Essential Services

When summoned by the Site Main Controller.

In-charge (Env. & Safety)

- Assists in evacuation of casualties and their removal to the medical centre.
- Mobilizes personal protective equipment and safety appliances and assists personnel handling emergency in using them.
- Assists male nurse in providing first aid to those who are injured.
- Keepsacheckonanynewdevelopmentofunsafesituationandreportthesametothe Incident Controller.
- Collects and preserves evidence to facilitate future inquiry.

- To boost morale of rescue team and worker.

Doctor / Male nurse

- Doctor/male nurse providing medical aid to the injured personnel and prepare case papers for those who are required to be admitted in a hospital.
- Call own ambulance &organize outside ambulance service to bring casualties to the medical center and to transport them to the hospitals.
- Arranges blood donors of various blood groups to assist hospitals in procuring blood.
- Contacts designated hospitals and nursing homes informing them about the emergency and alert them to be prepared to admit the injured.
- Carries out necessary formalities in the fatal cases.
- First aiders will assists the male nurse in B & C shift.

In-charge (Security & Fire) / Security cum Fire Officer / ACO

- Proceeds to fire station, summons fireman, establish contact with incident controller.
- To carry out firefighting under the direction of site main controller.
- Organizes outside assistance in firefighting if required
- Controls traffic movement, removes truck and tanker drivers outside the plant and prevents entry of all non-essential personnel.
- Cordons off the incident site and keeps the site clear of by unwanted persons.
- Organizes transportation of personnel from the assembly points to a safe location in or outside the plant.
- Organize outside assistance in firefighting and rescue operations if required.

Fire Warden

To assist fire response team in carrying out firefighting, rescue operation and duties such as cordon the area, traffic control etc. assigned by fire safety co-coordinator.

11. Support Services

When summoned by the Site Main Controller, the support services will be provided by all concerned HODs (like Mechanical, electrical, instrument &civil).

Mechanical

- Assumes charge for the smooth operations of all emergency equipment and systems.
- Takes charge of shift maintenance technicians and assigns them emergency repair work in consultation with Incident Controller.
- Makes available essential materials and spare parts from the stores.
- Makes available tools and tackles for rescue operation (e.g. mobile crane, wire/ nylon ropes, slings, D-Shackles, etc.)

Electrical

- Electrical isolation, additional connections, additional lighting & strengthening communication system.

Instrument

- Emergency operation of control system.

Civil

- To clear the access if require.
- Makes available tools and tackles for rescue operation (e.g. excavator, shovel, hammer, pneumatic barker, etc.)

HOD (HRD)

- Assumes charge of all external communication in consultation with Site Main Controller.
- Coordinates with Corporate Office and issues press or public statements if required.
- Ensure that medical aid is promptly provided to the casualties and their nominated person to contact in emergency is informed.
- Receives and meets important visitors and Government Officials.
- Arrange for canteen facility to rescue team / emergency team for drinking water & refreshment.

7.1.8 Emergency Evacuation

When an emergency occurs, it is necessary to evacuate personnel from the affected area of the plant who is not directly involved in the emergency operation. Depending upon the magnitude of the emergency, sometimes it is desirable to evacuate personnel from the adjacent area also if there is a possibility of the incident escalating affecting the area.

The necessity for evacuation in the plant will arise out of the intensity of thermal radiation due to fire in the affected area or due to toxic gas release which would also extend up to the adjacent area.

- For the purpose of evacuating personnel, safe assembly points are chosen and clearly marked. The assembly points are selected in the upwind direction / perpendicular to the wind direction so as to eliminate the downwind hazards due to fire and toxic release. We have an on line weather monitoring station. Wind direction & wind speed data is recorded& used in case of toxic release to facilitate evacuation and to know dispersion speed of chlorine.

Decision for evacuation of personnel will be taken by Incident Controller, who will ensure that the

- Person is to be assigned for each assembly point personnel to be evacuated are informed on public address system.
- Injured and trapped personnel are rescued by fire squad equipped with protective gear.
- Head count is taken by ACO with the help of senior most person available at the assembly point and a list of personnel evacuated is made.
- Instructions are given to vehicle drivers on the route to be followed and safe place where the personnel are to be taken.

7.1.9 Communication System

Communication is the lifeline of handling an emergency. When an emergency occurs it is necessary to raise the alarm immediately, to declare an emergency, to inform the emergency controllers, to inform the plant emergency services and affected areas within the plant as well as outside if necessary.

Therefore, the following facilities are provided in the plant for an effective two way communication:

- Intercoms, Public Address System, company mobiles and Willkie Talkie for normal and emergency in-plant communication.
- Fire and Emergency Sirens for raising an alarm for emergency and alerting all essential services.
- Non-Dedicated External Telephone for emergency contacts with Works emergency controllers, fire brigades, hospitals and police and for all other contacts including District Authorities, Government Agencies, neighboring Industries etc.

When the external telephones are not working, an emergency vehicle with a messenger could also be used for outside communication. Communication to the neighboring public, if necessary, should always be made through police and their wireless van.

Addition to the above, there will be siren system, which will be sounded for different stages of emergency.

7.1.10 Onsite Emergency Plan

Emergency procedures

Handling of an emergency calls for critical planning and ensuring a state of readiness at all times. Proper handling of the actual emergency needs certain actions to be taken before the emergency to ensure that all systems are ready and the emergency can be handled smoothly. Similarly after an emergency rehabilitation and reconstruction programmers are necessary. The actions to be taken before, during and after an emergency are described below:

Actions to be taken before the Emergency

1. Training programme should be conducted at regular intervals for
    - Use of Personnel Protective Equipments
    - Use of chlorine kit
    - Fire fighting
    - First Aid
2. Drills to be conducted for testing of emergency plans once in a six month.
3. Safe handling procedures (MSDS) for the chemicals handled in the plant should be written and copy made available in plant area.

Actions to be taken during the Emergency

For major emergency situations the various actions to be taken during emergency are as follows:

Emergency Response to Toxic Material Release

Among the hazardous chemicals handled at the facility, the worst emergency situation will involve release of chlorine gas.

A chlorine leak is easily detected due to its peculiar odour. As soon as a leak in piping or equipment handling chlorine is noticed, the immediate action should be to evacuate the downwind area and make efforts to isolate the source of supply.

Personnel performing such emergency operations should wear all personal protective equipment including air line or self-contained breathing apparatus.

Immediately on report of a leak, the Incident Controller should take following actions.

- Declare the EMERGENCY
- Evacuate the downwind area. If the consequences are likely to affect outside population, use available means to inform the public and nearby installations about the incident and possible effects and actions.
- Instruct to isolate the system supplying gas to source of leak and arrange safe shutdown of the plant.
- If emergency situation permits, attempt repair using necessary precautions.
- Perform all other responsibilities as per the Emergency Response Plan.

The following actions to be taken for the hazardous chemicals handled in the plant:

Chlorine

Since Chlorine is approximately 2.5 times heavier than air, it has a tendency to spread at lower levels. If the leakage is heavy the source can be immediately detected, by the pungent odour and colour of the gas.

- As soon as there is any indication of the presence of chlorine in the air immediate steps should be taken to remedy the situation. The leak source can be detected with the help of ammonia torch (a wooden stick with piece of cloth at one end. It is soaked in ammonia solution. When taken near the leak source, it will give dense white fumes.)Leakages, if any should be immediately reported to the supervisor.
- Leakage once detected, should be attended immediately, to avoid further deterioration of leakage.
- Use personal protective equipment's such as gas mask with canister, on line air breathing apparatus, self-contained B.A. set while entering the leakage zone.
- Leakage point should be preferably approached from higher elevation and upwind direction.
- Isolate the system as quickly as possible and evacuate the area. Only persons required for combating the leakage should be allowed to enter the area.
- Water should never be sprayed on a chlorine leak. To do so will make the leak worse because of the corrosion action of wet chlorine. Moreover, heat from water increases evaporation rate of chlorine.
- If the container is leaking in such a position that chlorine is escaping as liquid, the container should be turned so that chlorine gas escapes. This reduces the hazard tremendously due to the following reasons:
  - For a given size of hole and a given internal pressure, the rate of gas leak will be about $1/15^{th}$ of the liquid escape rate.
  - Volume of liquid chlorine = 460volume of chlorine gas
- The severity of a chlorine leak may be lessened by reducing the pressure on the leaking container. These may be done by absorbing chlorine gas (not the liquid) from the container in a solution of caustic soda, soda ash or hydrated lime.
- If there is leakage from the valve seat, valve body or valve inlet threads, use tie rod assembly from tonner safety kit to plug the leak.
- If there is hole in the container body, use chain assembly from tonner safety kit.
- If the leak cannot be plugged, reduce pressure in the container or remove it to an isolated area.

Hydrochloric acid

This is a highly corrosive liquid and causes burns on contact with skin. Utmost precaution should be taken while carrying out maintenance jobs or while attending leakages.

- Put on all required personal protective equipment's such as face mask/safety goggles, hand gloves, helmet, apron, gasmask, gumboots etc. while attending the leakages.
- Do not attempt to attend leakages on pressurized line/vessel or on running equipment.
- Small or large quantities of hydrochloric acid should be neutralized with caustic lye/lime.

Sulphuric acid

This is a highly corrosive and reactive liquid and causes severe burns on contact with skin. It attacks and corrodes many metals by releasing hydrogen. Utmost precaution should be taken while carrying out maintenance jobs or while attending jobs or while attending to leakages.

- Put on all required personnel protective equipment such as neck to toe overall face mask/safety goggles, hand gloves, apron and gum boots etc. while attending leakage or leakage point.
- Do not attempt to attend leakages on pressurized line/vessel or on running equipment.
- Small quantities of acid spillage may be diluted with large quantities of water. If the quantity of acid is large, do not use water to avoid generation of heat of solution. The solution should be neutralized with soda ash or with calcium hydroxide.

Caustic lye

This is a highly reactive liquid and causes burns on contact with skin. Utmost precaution should be taken while carrying out maintenance jobs or while attending leakages.

- Put on all required personal protective equipment's such as face mask / safety goggles, hand gloves, helmet, apron, gasmask, gum boots etc. while attending the leakages or approaching to leakage point.
- Do not attempt to attend leakages on pressurized line/vessel or on running equipment.
- A small quantity of caustic lye may be diluted with a large quantity of water.

Emergency Response to a Storage Tank Fire/ Coal fire / Explosion

Following possible approach may be used to handle an emergency situation involving a tank fire / coal fire / explosion:

1. Assesses the situation and gather required information on the accident including suspected casualties, rescue operations required, materials involved, tank details, pipe line / flange details, weather conditions and resources available.
2. Consider possible alternatives to bring the situations under control and make decisions with reference to line of action i.e. whether to attempt to extinguish the fire or only to control its spread, requirement of emergency response personnel or not etc.
3. Take appropriate response actions :
   - Isolate the area and remove all nonessential personnel.
   - Isolate / shut off the valve of source of fuel from up wind direction.
   - Switch off all power supply specifically temporary power supply.
   - To stop welding, gas cutting and all hot job in surrounding areas and specifically in down wind direction

- Evaluate the area downwind of the fire or possible vapor clouds.
- If remote isolation of the vessel is possible, provide for doing so, to avoid flammable vapor cloud formation.
- Attempt to extinguish the fire using the appropriate extinguishing agents and methods.
- Approach to fire from upwind. The personnel involved should wear appropriate protective equipment.
- Cool the vessel involved in fire and adjacent vessel with water to prevent spread of the fire.
- Prevent the overflowing of fire water from dykes and other containment systems.

Actions to be taken after the Emergency

After decision has been made by Site Main Controller to call off the emergency, following necessary actions are recommended:

- Analysis of the incident.
- Rehabilitation of plant and offsite personnel.
- Repairing the plant machinery and reconstruction of structure\
- Availability of On-site emergency plan

Copy of latest on-site emergency plan is available at following locations.

- Main plant control room
- Power plant control room
- Main gate security
- Safety Health & Environment department
- Technical services
- Occupational Health Center
- Mechanical department
- Instrumentation department
- Civil department
- Electrical department
- Quality Assurance department
- Purchase department
- HRD
- Accounts
- Sales co –ordination department
- Store department
- Information Technology Department

7.1.11 Off-Site Emergency Plan

Need of the off-site emergency plan

The off-site emergency plan prepared here in will deal with the incident I the onsite plan which has the potential to harm persons or the environment outside the boundary of the factory premises.

The most significant risk to outside areas is that associated with a large release of chlorine. Spread of its effected outside the works may require traffic control, evacuation, shelter arrangement.

Off-site emergency plan has been drawn up with a view to mobilize resources and integrate with district contingency plan for an effective system of command and control in combating the emergency.

Thus the main purpose of the off-site emergency plan is:

- To provide the local / district authorities, police, fire brigade, doctors, surrounding industries and the public, the basic information of risk and environment impact assessment and to appraise them of the consequences and the protection prevention measures and control plans and to seek their help to communicate with the public in case of major emergency;
- To assist the district authorities for preparing the off-site emergency plan for the district or particulate area and to organize rehearsal from time to time and initial corrective action based on the lesson learnt.

Structure of the off-site emergency plan

The off-site emergency plan should be integrated properly with the district contingency plan to tackle any kind of emergency. The site main controller will keep liaison for this purpose with the district authorities.

External telephone facilities from Shree Ganesh Fats to mutual aid units shall be established for quick communication.

The names of the key persons shall be defined to establish contacts and Co-ordinate the activities with the help of the collectorate and disaster management center in case of major emergency.

An on-site emergency control room shall be identified which can be activated / used for emergency control and manned round the clock.

As far as off-site emergencies are concerned, information shall be received first by the police control room, next information to fire brigade and to disaster management center. The police / fire brigade control room shall in turn inform police commissioner, collector and Municipal Commissioner.

Arrangement made for off- site emergency

Considering distance from district Head Quarters and other nearby external emergency control organization, SGF shall make following arrangements in consultation with factory inspectorate, district collectorate.

1. **Mutual aid arrangements**
   To help each other during emergency situation the company shall enter to arrangement with neighboring industries.

2. **Disclosure of information to neighboring organization and population**
   SGF shall prepare booklet and circulate among neighboring organization and population dealing with hazardous operation and chemicals. This booklet shall include information about:

- First aid
- Emergency treatment
- Probable type of emergencies that can arise
- Preventive steps taken to control emergency
- Emergency warning siren code system, to make them aware in advance.

SGF shall also carry out group get together, acquaintance round, meeting with neighboring public and population to train, brief and make them aware about the operation and preparedness. The same group along with external emergency control organization shall be invited during mock drill, rehearsals for training and acquaintance.

7.1.12 Training Rehearsal & Records

Need for Training and Rehearsal

It is important that every emergency plan, when finalized should be written down and communicated to all concerned. After making people aware of such plants, the plan itself should be put to test, preferably in parts, to begin with, so that the effectiveness can be assessed and alternatives can be developed. After this, more extensive exercise can be conducted once a year involving outside agencies also. This will not only enable the industrial plant personnel to enhance their speed of mobilization, but also will help the outside agencies to improve their effectiveness in responding to emergencies.

These exercises will help to refine the procedures by identifying deficiencies and difficulties. At this stage more elaborate exercise can be planned to involve outside services who should be closely involved at the planning stage. Each exercise should be monitored by a member of independent observers located at various positions.

Mock drill records & Observation

SGF shall carry out mock drill and record of that shall be maintained, observation made during each mock drill shall be recorded and corrective action shall be taken to any short fall in action plan. Emergency plan shall be updated based on these short comings identified during the rehearsal.

7.1.13 Occupational Health

Well-equipped occupational health center manned by qualified MBBS doctor, Certified in Industrial Health and four male nurses having qualified certificate in Occupational Health, who shall work round the clock. The work of occupational health center includes:

- For new employees: Pre- employment medical checkup.
- Six-monthly health check-ups of all the employees (periodic medical examination).
- Physical examinations such as height, weight, chest, blood pressure, eye vision & skin test
- Lab test such as hemoglobin (TC, DC & ES tests), urine – complete tests (physical examination, macro test, micro tests like RBC, WBC, crystals, albumin, globulin, sugar)
- Spirometry test (lung function) for pulmonary impairment detection.
- Audiometry test – for the employees exposed to noise.
- Counseling with employees for improvement in health conditions.
- Medical check-up carried out to study the effects of vibration.
- Regular health survey is carried out with specific objective to address physical, biological and ergonomic issues of employees.
- Workplace monitoring Survey
- Health assessment of Employees after illness/injury

7.1.14 Disaster Management Plan

DMP for Chlorine

- On-site emergency plan in place and regular mock drills shall be conducted
- One liquid chlorine storage tank shall always be kept empty for emergency evacuation
- Double safety valves on each liquid chlorine storage tank shall be provided
- Remote isolation valves
- Emergency Chlorine neutralization system in place
- Customer training on Chlorine emergency
- Chlorine emergency team for handling eventuality outside factory

Provision of online chlorine sensors at various locations and connected with DCS is given in ***Table 7-4***

Table 7-4: Online Chlorine sensors

S. No	Location
1	Cl_2 storage tanks
2	Cl_2 bottling
3	HCl plant
4	Hypo plant
5	Cell house
6	Near Chlorine Compressor area

Safety System for Chlorine Storage and Handling

- Preventive inspection and maintenance
- Leakage collection and absorption system.
- Level indicators with alarm for liquid Chlorine storage tanks.
- Load cell for each Chlorine storage tank.
- Chlorine gas detector to detect leakage of Chlorine.
- Suction hoods.
- Rupture disc provided on each storage tanks
- Two relief valves are provided on each of the storage tanks
- Remote operated auto valve.
- Storage system - approved by CCE

Safety System for Chlorine Bottling and Handling

- Weighing arrangement shall be provided in each EOT crane to check the weight of receiving / dispatch tonners.
- Degassing of tonner shall be done in Sodium Hypochlorite plant.
- Hydro testing – Every two years / need basis.
- Hydro testing tonners shall be segregated separately. If wt. loss is more than 5% it shall be rejected
- Auto filling system shall be provided.

- Load cell shall be provided on each filling post to measure the weight & calibration of load cells shall be done on daily basis.
- Before dispatch of tonners, it shall be observed for one day for any abnormality.
- Valves shall be covered with metallic cover hood.
- Chlorine sensors shall be installed. These sensors shall be connected with DCS.
- Suction hood covering complete tonners shall be provided.
- Tonner tracking system

DMP for Hydrogen

- On-line monitoring of Hydrogen in Chlorine
- Flame arrester on Hydrogen gas line
- Electrical appliances in Hydrogen compressors area shall be flameproof.
- Hydrogen gas detectors shall be installed at Hydrogen compressor and activates alarm in case of Hydrogen leakage
- Fire hydrant system throughout the plant.
- Water Sprinkler shall be provided around transformers
- Round the clock availability of Fire Tender
- Mutual aid schemes with the adjoining industries
- Product storage tanks provided with dyke wall
- Four levels of communication (Telephone, P A system, Willkie talkie, Siren system)
- Incident investigation and corrective system including near-miss cases.

Brief description of the safety measures taken for Hydrogen storage

- Preventive inspection and maintenance
- Flame proof electrical appliances, tools
- Earthing strip at entry to discharge electrostatic charge
- Earthing from both sides of the trucks during filling & being followed
- Hydrogen sensors with alarm for immediate detection on all hydrogen compressor
- Jumpers on all the flange joints
- Storage system

Safety System for Coal storage and Handling

- Storage area isolated by providing RCC wall
- Water sprinkler system
- Fire hydrant system
- Fire tender with firefighting crew
- Water spraying system
- Prevention of air movement through coal stacks by compacting by mechanical means
- Fire water reservoir

Safety System for Caustic pipeline

- Use of schedule 40, CS seamless pipeline.
- Pipeline laid above ground on firm & fixed supports.
- Pipe-line duly hydro-tested at 1.5 times the operating pressure.
- Fixed schedule for the regular inspection and maintenance of this piping.

Safety System for HCl line

- Use of FRB pipeline.
- Pipeline laid above ground on firm & fixed supports.
- Pipe-line duly hydro-tested at 1.5 times the operating pressure.
- Fixed schedule for the regular inspection and maintenance of this piping.